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# Factors Associated with a Family's Delay of Decision for Organ Donation After Brain Death

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**Background:** This study aimed to explore the factors associated with a family's delay of decision for organ donation after brain death, and to investigate the effect of such a delay on organ donation.

**Material/Methods:** Medical records and data on counseling about organ donation with the families of 107 brain-dead potential donors between September 2012 and March 2016 at a single tertiary medical center were retrospectively reviewed.

**Results:** The final consent rate was 58% (62/107), and successful donation was performed in 40% (43/107). Ninety-two families (86%) made a decision within 48 hours, whereas 15 (14%) required more than 48 hours for a final decision. In univariate and multivariate analyses, the independent factors associated with a decision delay were mean arterial pressure  $\leq 60$  mm Hg and coma therapy. In the early decision group ( $<48$  hours), the consent and successful donation rates were 55% (51/92) and 39% (36/92), respectively, whereas in the delayed decision group ( $\geq 48$  hours), these rates were 73% (11/15) and 47% (7/15), respectively. The consent and successful donation rates were not inferior in the delayed decision group.

**Conclusions:** These findings justify continuous efforts to maintain organ viability and to extend counseling to encourage donation even if the family cannot decide immediately.

**MeSH Keywords:** Brain Death • Tissue and Organ Procurement • Tissue Donors

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## Background

Patients who have been declared dead using neurologic criteria (irreversible loss of all brain function but maintained on a ventilator) are the single largest source of transplantable organs. However, only a small portion of these brain-dead donors are being used for solid organ transplantation, primarily because of the low percentage of families who consent to donation [1]. Numerous studies have been conducted on the factors affecting such consent [1–4].

Usually the decision of whether or not to donate is forced on family members in a complex clinical situation. It can be difficult to accept the death of a loved one, and many family members are not prepared to understand the medical aspect of brain death because of emotional stress. In addition, one of the most stressful situations is when a family member has to make a decision about the loved one's body without his or her previously specified opinion about organ donation. Sometimes, these factors negatively affect the decision to donate and delay the final decision making. Such a delay, although justifiable, can be associated with refusal to donate or failure of successful donation. However, the factors associated with a family's delay of decision and the effect of such a delay have not been well studied in the literature.

Therefore, the goals of this study were to assess the factors associated with a family's delay of decision for organ donation after brain death, and to investigate the effect of such a delay on the final decision and success of organ donation.

## Material and Methods

Medical records and data on counseling about organ donation with the families of brain-dead potential donors between September 2012 and March 2016 at a single tertiary medical center were retrospectively reviewed.

The institution where this study was performed has been implementing a donor action program (DAP) since September 2012. According to the DAP, all potential organ donors fulfilling 3 criteria (identified brain injury, dependent on a ventilator, and Glasgow Coma Scale score <4) were reported to the coordinator of the Organ Procurement Organization (OPO) by the head nurses of the intensive care unit. Then, the OPO coordinator reviewed the patient's medical records to determine the suitability for organ donation, and the decision of whether or not to approach the family about organ donation was made primarily by the patient's medical attendant. When the medical attendant allowed a family approach, the OPO coordinator performed an in-depth discussion with the family. If the family consented to donation, prerequisite medical tests

were performed. In our country, legally, investigations of brain death including an apnea test and a test for brain stem reflex must be performed twice, and in adults the time interval between the primary and secondary investigations should be more than 6 hours. Confirmation of a flat electroencephalogram for more than 30 minutes is also indispensable. The final step is approval of the committee after examination of the entire process and legal review. In cases receiving coma therapy, the primary investigation must be postponed for more than 48 to 72 hours after cessation of coma therapy because measurement of serum barbiturate level is not available in our country. During coma therapy, non-invasive treatments for maintaining essential hemodynamics including balancing volume status, using a pressor agent, mechanical ventilatory support, nutritional support, treatment of infections, and lowering intracranial pressure were usually used. Regardless of whether coma therapy was ongoing or ended, if the medical attendant judged the patient was neurologically hopeless and agreed to start a conversation about organ donation, the OPO coordinator approached the families. The donation after circulatory death (DCD) program is not established yet in our country.

A total of 146 brain-dead potential donors were identified by the OPO coordinator, all of whom showed features of brain death, including comatose mental status, loss of self-respiration, pupil dilatation, and loss of brain stem reflex. Among them, 112 families were approached for organ donation. Five patients were excluded because they did not fulfill the criteria for diagnosis of brain death. Finally, 107 brain-dead potential donors were included in the analysis. Among these potential donors, no one was registered in the donor registry of our country.

Potential donors were categorized according to the time interval between the initial counseling and the final decision. The early decision group was defined as those with a time interval of fewer than 48 hours, while the delayed decision group was defined as those with a time interval of more than 48 hours. Clinical data on time of donor identification were collected retrospectively. Factors potentially affecting a family's delay of decision (including age, sex, cause of brain injury, need for inotropic drugs, vital signs, treatment method, serum sodium level, and family relationship) were analyzed, and clinical courses between groups were compared.

Univariate analysis of categorical variables was performed using the chi-square test or Fisher's exact test. Then, variables determined to be statistically significant in the univariate analysis were used in subsequent multivariate analysis, which used a logistic regression model with binary variables.

## Results

The baseline characteristics of the 107 potential donors at the time of identification are summarized in Table 1. The final consent rate for donation was 58% (62/107), and successful donation was performed in 40% (43/107). Of 107 families, 92 (86%) were categorized in the early decision group (decision time <48 hours), whereas 15 (14%) were categorized in the delayed decision group (decision time  $\geq$ 48 hours). Characteristics including age, sex, cause of brain injury, need for inotropic drugs, blood pressure, craniotomy or coma therapy as a treatment, serum sodium concentration, and prior determiner among the family members were investigated. In univariate analysis, mean arterial pressure  $\leq$ 60 mm Hg ( $p=0.046$ ) and coma therapy ( $p=0.011$ ) were associated with a delay of decision (Table 2). In multivariate analysis, these 2 factors were found to be significant independent predictors (mean arterial pressure  $\leq$ 60 mm Hg:  $p=0.029$ ; odds ratio, 5.245; 95% confidence interval, 1.186–23.194; coma therapy:  $p=0.014$ ; odds ratio, 4.575; 95% confidence interval, 1.363–15.350) (Table 3).

In the early decision group, the consent rate was 55% (51/92), and successful donation was performed in 71% (36/51) of consenting donors, whereas in the delayed decision group, these rates were 73% (11/15) and 64% (7/11), respectively. The consent and successful donation rates were not significantly different between groups (Table 4). In fact, the final consent rate was slightly higher in the delayed decision group than in the early decision group (73% vs. 55%); however, the difference was not significant ( $p=0.263$ ). The clinical courses of both groups are illustrated in Figure 1.

In 19 potential donors whose families consented to donation, the donation process could not be completed successfully. Rapid clinical aggravation was the most common cause (10/19, 53%). Clinical aggravation during the waiting period after cessation of coma therapy (6/19, 32%), consent provided too late (2/19, 11%), and uncertainty of cause of brain death (1/19, 5%) were other causes of donation failure.

## Discussion

The decision for organ donation is often made in complex circumstances. Emotional upheaval in acutely bereaved families and lack of clarity regarding brain death engender dissonance and distress that adversely affect a family's decision making and grieving over time [5–7]. Numerous factors have been shown to affect the decision of family members [1,2,4]. Research in the United States has cited emotional exhaustion and inadequate staff sensitivity and compassion to address the family's suffering among a number of reasons for not consenting to donation [8–10]. The complex situation and

emotional stress make it difficult for families to understand the nature of brain death and to accept the real death of their loved one, which ultimately disturbs the best decision regarding organ donation. As a result, a final decision might require several hours to several days, and such a delay can negatively influence the consent rate. However, the factors associated with a family's delay of decision and the effect of such a delay have not been well studied. This study is the first to focus on this subject and take delay of decision into consideration as a factor that affects consent and successful donation.

The consent rate for organ donation in Europe is known to be 50% to 80% [2,4]. Several studies have found that approximately 85% of families are asked to donate, but only 50% provide consent. Other studies have confirmed these findings [11–13]. However, most of these reports regarding consent for organ donation after neurologic death are based on Western populations. The practical consent rate in Asian countries is not well known. Traditionally, in Asian cultures, especially Korean, Japanese, and Chinese, the body of a loved one should not be injured after death, which is based on a belief originating from the Confucian tradition. This has been believed to be the main cause of low consent for organ donation in Asian countries, without concrete evidence supporting it. In this study, the consent rate among those who were approached was 58%, which is comparable with reports from Western countries. This finding might indicate that Asian cultures have changed quickly during the past decades, and as a consequence, the general attitude about organ donation also has changed. Large-scale research in Asia is required about this subject.

Families of potential donors with mean arterial pressure  $\leq$ 60 mm Hg or receiving coma therapy seemed to delay a final decision for more than 48 hours. Thus, these conditions appear to be possible important obstacles to successful organ donation. Potential donors with these conditions are at high risk of circulatory death while waiting for the family's final decision. The psychological etiologies of how these conditions affect a delay of decision were not verified in this study. If the patient's vital signs are already unstable and the families are expecting the impending circulatory death, the primary decision maker among the family members might want to avoid additional physical injury to the loved one and want to end the grievous situation. Families of potential donors receiving coma therapy might expect possible clinical improvement with this therapy and might want to delay a final decision. Despite the uncertainty about how these situations affect the final decision, we can draw a careful conclusion that the medical attendant should hasten the counseling with the OPO coordinator as soon as possible so as not to lose an opportunity for organ donation.

In this study, the final consent and successful donation rates were not significantly different according to the decision

**Table 1.** Characteristics at the time of potential donor identification (N=107).

| Variables                                    | Characteristic details          |
|----------------------------------------------|---------------------------------|
| Age, mean                                    | 49.5±15.7 years                 |
| Range                                        | 3–83 years                      |
| Age ≥65 year, No. (%)                        | 19 (18)                         |
| Age <65 year, No. (%)                        | 88 (82)                         |
| Sex, No. (%)                                 | Male 37 (35); Femal 70 (65)     |
| Cause of brain death                         |                                 |
| Cerebrovascular accident, No. (%)            | Hemorrhage 44 (41)              |
| Trauma-related, No. (%)                      | Infarct 5 (5)                   |
| Hypoxic brain injury, No. (%)                | Head trauma 25 (23)             |
| Etc, No. (%)                                 | Post-CPR 11 (10)                |
|                                              | Hanging 9 (8)                   |
|                                              | Hypoxic, etc 9 (8)              |
|                                              | Brain tumor 3 (3)               |
|                                              | Ventriculitis 1 (1)             |
| Feature of brain death                       |                                 |
| Mental status, No. (%)                       | Coma 107 (100); Non-coma – (0)  |
| Loss of self-respiration, No. (%)            | Yes 107 (100); No – (0)         |
| Pupil dilatation, No. (%)                    | Yes 107 (100); No – (0)         |
| Loss of brain stem reflex, No. (%)           | Yes 107 (100); No – (0)         |
| Inotropics, No. (%)                          | Yes 93 (87); No 14 (13)         |
| SBP, mean                                    | 113.6±23.7 mmHg                 |
| Range                                        | 54–171 mmHg                     |
| SBP >90 mmHg, No. (%)                        | 90 (84)                         |
| SBP ≤90 mmHg, No. (%)                        | 17 (16)                         |
| MAP, mean                                    | 83.8±17.6 mmHg                  |
| Range                                        | 44–139 mmHg                     |
| MAP >60 mmHg, No. (%)                        | 96 (90)                         |
| MAP ≤60 mmHg, No. (%)                        | 11 (10)                         |
| Craniotomy, No. (%)                          | Yes 38 (36); No 69 (65)         |
| Serum Na <sup>+</sup> , mean                 | 150.3±9.8 mEq                   |
| Range                                        | 128–170 mEq                     |
| Serum Na <sup>+</sup> ≥160 mEq, No. (%)      | 22 (21)                         |
| Serum Na <sup>+</sup> <160 mEq, No. (%)      | 85 (79)                         |
| Coma therapy, No. (%)                        | Yes 40 (37); No 60 (63)         |
| Primary decision maker among family members  |                                 |
| Parent, No. (%)                              | 21 (20)                         |
| Descendant, No. (%)                          | 21 (20)                         |
| Spouse, No. (%)                              | 60 (56)                         |
| Sibling, No. (%)                             | 5 (5)                           |
| Final decision about organ donation, No. (%) | Consent 62 (58); Refuse 45 (42) |

CPR – cardiopulmonary resuscitation; SBP – systolic blood pressure; MAP – mean arterial pressure.

duration. In fact, the final consent rate was higher in the delayed decision group than in the early decision group (73% vs. 55%,  $p=0.263$ ) (Table 4). This result may indicate that the delay of decision does not reflect a negative attitude about organ donation, but rather, a reasonably necessary time for deliberate decision. Therefore, the medical attendant and OPO

coordinator should continue the efforts to maintain organ viability and consider extended, repetitive counseling to encourage donation. However, it is worthwhile to note that donation failure despite the family's consent was relatively higher in the delayed decision group (27% vs. 16%,  $p=0.464$ ) (Table 4).

**Table 2.** Factors affecting decision delay  $\geq 48$  hours (univariate analysis).

| Variables                            | Decision time <48 hours<br>(N=92) | Decision time $\geq 48$ hours<br>(N=15) | P-value            |
|--------------------------------------|-----------------------------------|-----------------------------------------|--------------------|
| Age $\geq 65$ years                  | 19 (21%)                          | (0%)                                    | 0.068 <sup>#</sup> |
| Female sex                           | 61 (66%)                          | 9 (60%)                                 | 0.634*             |
| Post-CPR hypoxic brain injury        | 8 (9%)                            | 3 (20%)                                 | 0.183 <sup>#</sup> |
| Inotropics                           | 80 (87%)                          | 13 (87%)                                | 1.000 <sup>#</sup> |
| SBP $\leq 90$ mmHg                   | 13 (14%)                          | 4 (27%)                                 | 0.253 <sup>#</sup> |
| MAP $\leq 60$ mmHg                   | 7 (8%)                            | 4 (27%)                                 | 0.046 <sup>#</sup> |
| Craniotomy                           | 31 (34%)                          | 7 (47%)                                 | 0.330*             |
| Serum Na <sup>+</sup> $\geq 160$ mEq | 17 (19%)                          | 5 (33%)                                 | 0.187*             |
| Coma therapy                         | 30 (33%)                          | 10 (67%)                                | 0.011*             |
| Spouse as primary decision maker     | 54 (59%)                          | 6 (40%)                                 | 0.176*             |

\* Chi-square test; <sup>#</sup> Fisher's exact test.

**Table 3.** Factors affecting decision delay  $\geq 48$  hours (multivariate analysis\*).

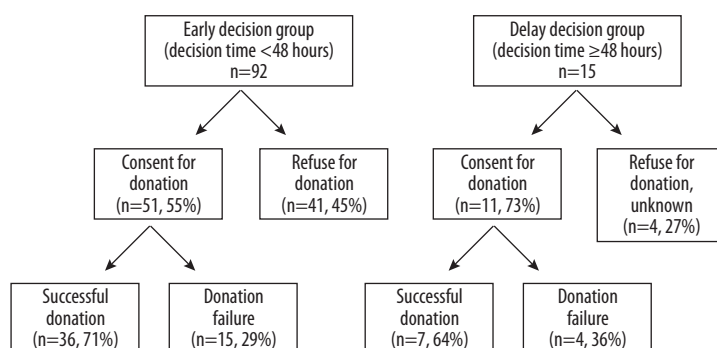
| Variables          | P-value | Odd ratio (95% C.I.) |
|--------------------|---------|----------------------|
| MAP $\leq 60$ mmHg | 0.029   | 5.245 (1.186–23.194) |
| Coma therapy       | 0.014   | 4.575 (1.363–15.350) |

\* Binary logistic regression model.

**Table 4.** Effects of decision time on organ donation.

|                                         | Decision time <48 hours<br>(n=92) | Decision time $\geq 48$ hours<br>(n=15) | P-value            |
|-----------------------------------------|-----------------------------------|-----------------------------------------|--------------------|
| Final decision – consent                | 51 (55%)                          | 11 (73%)                                | 0.263 <sup>#</sup> |
| Successful donation                     | 36 (39%)                          | 7 (47%)                                 | 0.581*             |
| Donation failure after family's consent | 15 (16%)                          | 4 (27%)                                 | 0.464 <sup>#</sup> |

\* Chi-square test; <sup>#</sup> Fisher's exact test.



**Figure 1.** Clinical course according to decision time.

The principle of decoupling is a well-known way to increase the consent rate for organ donation [14]. A study conducted by Kentucky Organ Donor Affiliates in 1989 to 1990 reported that the consent rate increased from 18% to 60% if there was a separation of time between the pronouncement of death and the approach for organ donation [14]. Our institution also follows the decoupling principle. However, when the hemodynamics of a potential donor worsen, the ideal practice of decoupling frequently becomes impossible. The patient's attending physician may feel an ethical conflict about providing active or invasive life-support care that seems to have no therapeutic benefit for the patient's recovery and appears to have significance solely for maintaining organ quality, especially when the family's opinion about organ donation is not specified yet. This special situation involving insufficient time between the pronouncement of death and communication on organ donation frequently occurs in the emergency department [15]. Sometimes, the ideal principle of decoupling is not consistent with the current recommendation of early referral to the OPO coordinator [15]. If there is value in organ donation, which we profoundly believe there is, a more flexible, high-dimensional strategy is required when a delay of decision is expected and the potential donor is progressing to circulatory death. In this study, the most common (53%) cause of donation failure despite the family's consent was rapid aggravation of hemodynamics of the potential donor. This leaves a little room to increase successful donation by early referral to the OPO coordinator and requires an active strategy for an anticipated delay of decision.

In this study, the OPO coordinator was not allowed to provide counseling about organ donation to 34 families (23%) of 146 potential donors. It is well known that involvement of a specialist nurse is a strong factor for family consent [1,2,4]. Various clinical situations can prohibit a family approach [2]. Although the detailed reasons for the inability to counsel were not verified in this study, the fear of attending physicians about starting the conversation about organ donation might be one of the causes. However, a randomized, double-blind, controlled trial on the benefits of organ donation in the community reported

that health care staff can present information about the donation process and the benefits of organ donation and transplantation without the risk of pressuring families into providing consent against the wishes of their loved one [16].

We acknowledge a number of limitations to our study. First, there is a lack of detailed information about the factors that could affect the family decision making. Most importantly, our data did not include the previously mentioned attitude of the potential donors, or the religious background or socioeconomic status of the families. In addition, data on ethnicity, the circumstances in which the referral was made, the timing of the formal family approach, and the details of the conversation are lacking. These factors have been analyzed in previous studies [1–4]. Second, a significant portion of the identified potential donors were excluded from a family approach. The consent rate therefore may not reflect the status of real world. Third, since a DCD program was not established in our country, the failure of organ donation in spite of families' consent was relatively common. Because different countries have different systems and laws regarding organ donation, the findings in this study cannot be extended around the world.

## Conclusions

A family's delay of a decision for organ donation is common (14%) and may negatively influence successful organ donation. In potential brain-dead donors with mean arterial pressure  $\leq 60$  mm Hg or receiving coma therapy, early referral to the OPO coordinator is recommended. In this study, the consent and successful donation rates were not inferior regardless of a decision delay of more than 48 hours. Therefore, continuous efforts to maintain organ quality and extend counseling to encourage donation are justified even if the family cannot decide immediately.

## Conflicts of interest

The authors have declared no conflicts of interest.

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